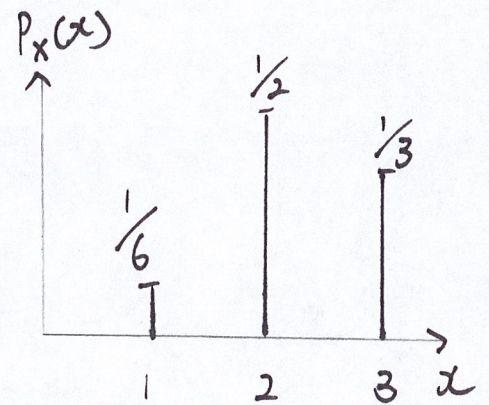


Expectation

III Motivating example : Play a game.

let $X = \$$ of reward receiving at each play.



Q: How much is expected to win in each play if the game is played many times.

A: Expected \$ of reward

$$= \frac{1}{6} \times 1 + \frac{1}{2} \times 2 + \frac{1}{3} \times 3$$

$$= 2$$

Def. ... The expectation of a RV X is

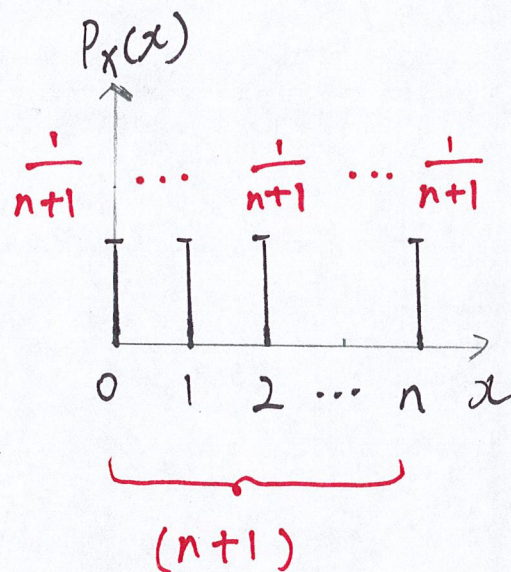
$$E(X) = \begin{cases} \sum_{\forall x} x \cdot p_X(x) & , \text{ discrete} \\ \int_{-\infty}^{\infty} x \cdot f_X(x) dx & , \text{ continuous} \end{cases}$$

Interpretations of $E(X)$:

- **Frequency**: the average in a large # of repetition of the same experiment.
- **Geometry**: the center of gravity of the PMF or PDF.

Ex. Let X be a RV distributed uniformly over $0, 1, 2, \dots, n$.

Find $E(X)$.



Approach 1.

$$E(X) = \sum_{x=0}^n x \cdot P_X(x)$$

$$= \sum_{x=0}^n x \cdot \frac{1}{(n+1)}$$

$$= \frac{1}{(n+1)} \cdot \sum_{x=0}^n x$$

Arithmetic sequence

$$= \frac{\begin{matrix} \text{first term} \\ \downarrow \end{matrix} (0 + \begin{matrix} \text{last term} \\ \downarrow \end{matrix} n) \cdot \begin{matrix} \text{\# of terms} \\ \uparrow \end{matrix} (n+1)}{2}$$

$$\times \frac{1}{(n+1)}$$

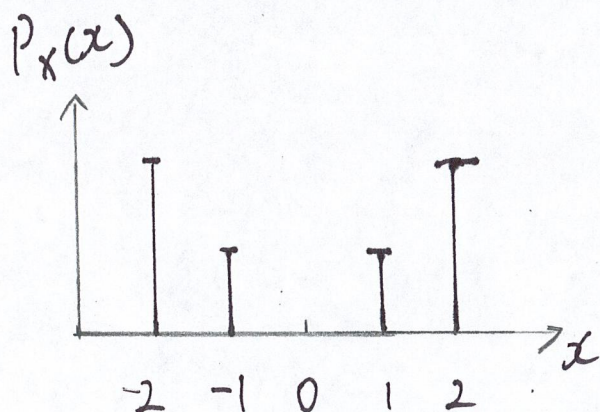
$$= \frac{n}{2}$$

Approach 2.

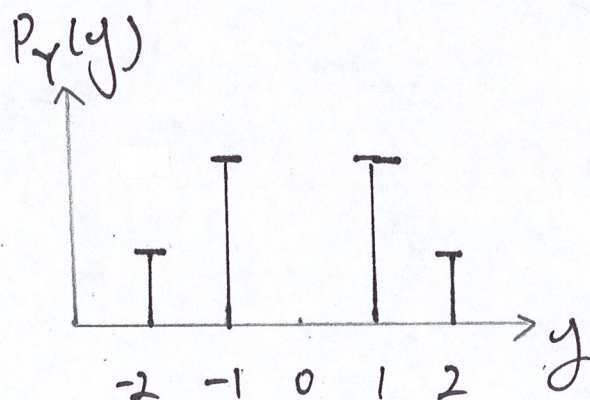
$E(X)$ = the center of gravity of $P_X(x)$

$$= \frac{n}{2}.$$

NOTE: Consider 2 RV's X and Y :



$$E(X) = 0$$



$$E(Y) = 0$$

$$\therefore E(X) = E(Y) \quad \not\Rightarrow \quad P_X = P_Y$$

$$E(X) = E(Y) \quad \Leftarrow \quad P_X = P_Y$$